Hardy-Weinberg Equilibrium

(1908)- Hardy in England and Weinberg in Germany demonstrated that gene frequencies do not depend upon dominance or recessiveness but remain essentially unchanged from generation to generation.

- Whatever the initial genotype frequences for two autosomal allele may be, after one generation of random mating, the genotype frequences will be $p^2:2pq:q^2$

- Both these genotype frequences and the allele frequences will remain constant in succeeding generations, unless factors not yet considered should change them.

Hardy-Weinberg Equilibrium

For equilibrium to work: Assume

- Random mating: Every individual has an equal chance of mating.
- The population is infinitely large: this ensures that there is no chance deviation of the allele or genotype frequencies due to chance event (random genetic drift)
- No gene flow or migration: no genes are added into the population or removed from the population.
- All individuals have equal chances for survival and reproduction.
- No mutations

Hardy-Weinberg Equilibrium

Population: A group of sexually interbreeding or potentially interbreeding individuals. The population is usually considered as a local group (deme), in which each member has an equal chance of mating with any member of the opposite sex.

Populations have two important attributes – gene (or allele) frequencies and a gene pool.

The **gene pool** is the sum total of genes in the reproductive gametes of a population. It can be considered as a gametic pool from which samples are drawn at random to form the zygotes of the next generation.

- If **p** and **q** are the allele frequencies, the expected genotype frequencies are **p^2, 2pq, and q2**.

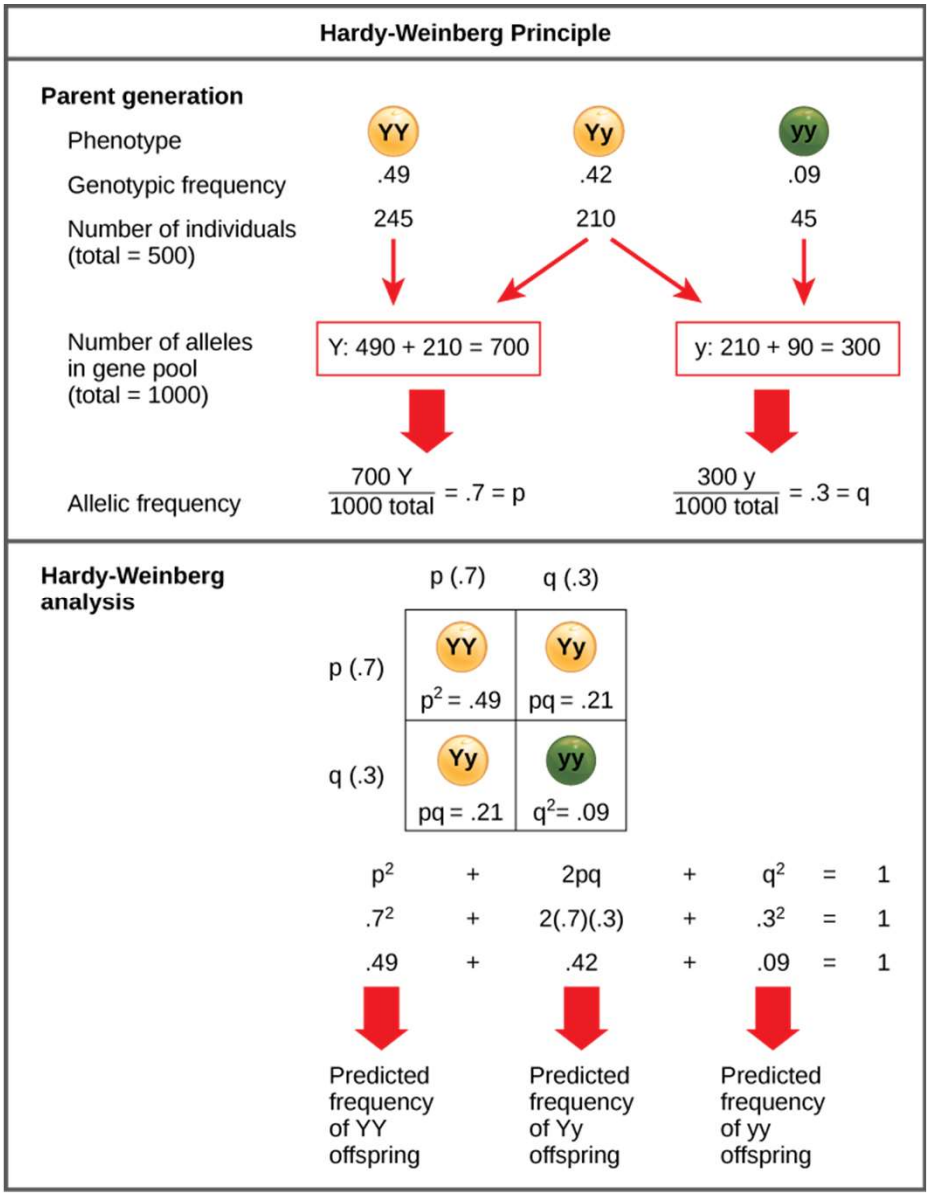
$$p^2 + 2pq + q^2 = 1$$

This is possible only when the population is under Hardy – Weinberg equilibrium. In other words when these five assumptions are met.

sperm

		A (p)	a (q)
egg	A (p)	AA (p^2)	Aa (pq)
	a (q)	Aa (pq)	aa (q^2)

- Hardy-Weinberg principle of equilibrium, states that a population's allele and genotype frequencies are inherently stable— unless some kind of evolutionary force is acting upon the population, neither the allele nor the genotypic frequencies would change.


Hardy-Weinberg Principle
Parent generation
Phenotype
YY
Yy
yy
Genotypic frequency
.49
.42
.09
Number of individuals (total = 500)
245
210
45
Number of alleles in gene pool (total = 1000)
Y: 490 + 210 = 700
y: 210 + 90 = 300
Allelic frequency
700 Y / 1000 total = .7 = p
300 y / 1000 total = .3 = q
Hardy-Weinberg analysis
p (.7)
q (.3)
YY
Yy
yy
p² = .49
pq = .21
pq = .21
q²= .09
p² + 2pq + q² = 1
.7² + 2(.7)(.3) + .3² = 1
.49 + .42 + .09 = 1
Predicted frequency of YY offspring
Predicted frequency of Yy offspring
Predicted frequency of yy offspring

Hardy-Weinberg frequencies of genotypes AA, Aa, and aa in relation to the frequency of the gene a.

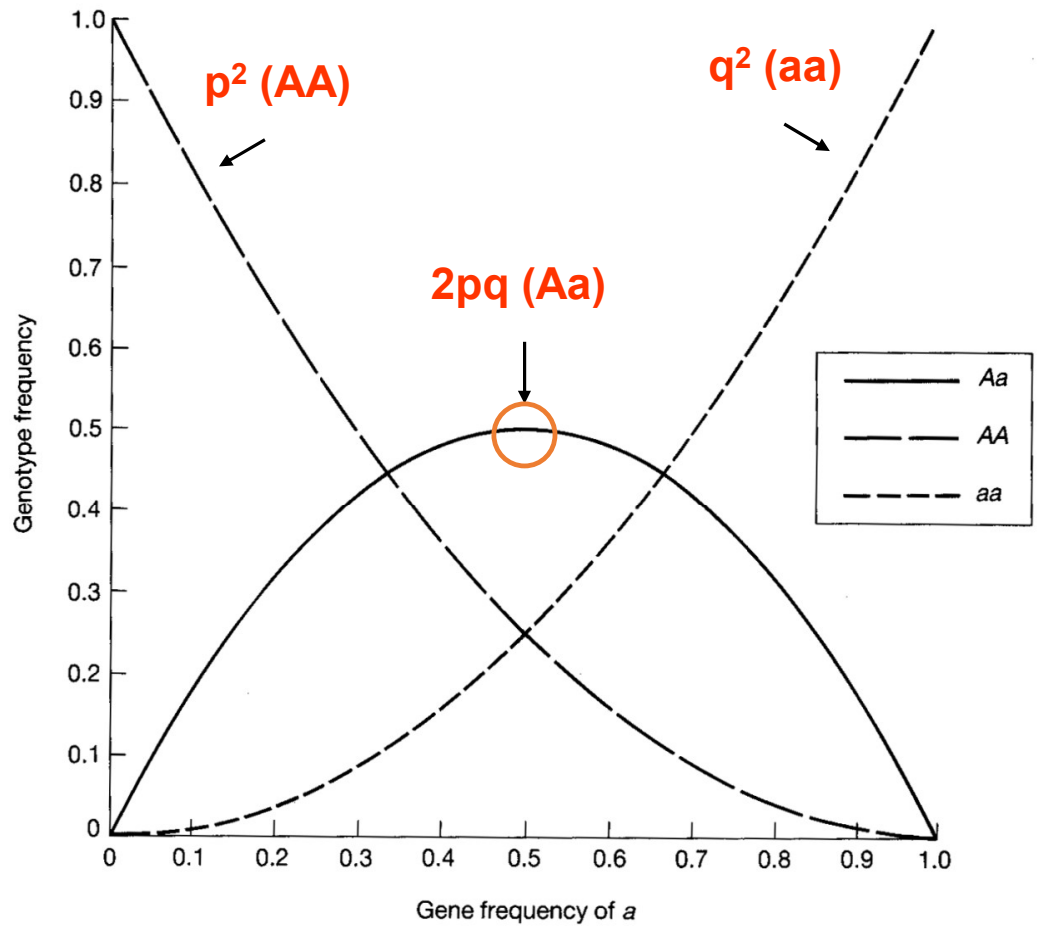


Figure 5.2 Hardy–Weinberg proportions of genotypes *AA*, *Aa*, and *aa* in relation to the frequency of the gene *a* (*q*)

We can test, by observation, whether genotypes in a population are in H-W equilibrium.

Significance of Hardy Weinberg principle

- Populations in nature are constantly changing in genetic makeup due to drift, mutation, possibly migration, and selection.

- So, the only way to determine the exact distribution of phenotypes in a population is to go out and count them.

- However, the Hardy-Weinberg principle gives scientists a mathematical baseline of a non-evolving population to which they can compare evolving populations and thereby infer what evolutionary forces might be at play.

- If the frequencies of alleles or genotypes deviate from the value expected from the Hardy-Weinberg equation, then the population is evolving.

PRIMARY USES OF THE H-W PRINCIPLE:

1) Enables us to compute genotype frequencies from generation to generation, even with selection.

2) Serves as a **null** model in tests for natural selection, nonrandom mating, etc., by comparing observed to expected genotype frequencies.

3) Expected heterozygosity provides a useful means of summarizing the molecular genetic diversity in natural populations.

Genotype frequencies for Wing-color Polymorphism in a Natural Population of the Moth *Panaxia dominula* (Fisher & Ford, 1947)

Color Pattern	*dominula*	*medionigra*	*bimaculata*	Total
Genotype	B_1B_1	B_1B_2	B_2B_2	
Sample Size (N_{ij})	905	78	3	N = 986
Frequency (P_{ij})	0.918	0.079	0.003	1.000

Allele Frequencies:

$p_1 = p_{11} + ½\, p_{12} = 0.918 + ½\,(0.079) = 0.958$
$p_2 = 1 – p_1 = 0.041$

Expectations ($\hat{N}_{ij}$)	$p_1^2 N$	$2p_1p_2 N$	$p_2^2 N$	
	905	79	2	

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EVOLUTIONARY THOUGHT AFTER DARWIN

- By the 1870s, most scientists accepted the historical reality of evolution (and this has been true ever since).
- It would be at least 60 years after the publication of *The Origin of Species* before natural selection would come to be widely accepted.
- People wanted life itself to be purposeful and creative, and consequently did not find natural selection appealing.

  - Neo-Lamarckism -- inheritance of acquired characteristics.
  - Orthogenesis -- variation that arises is directed toward a goal.
  - Mutationism -- discrete variations are all that matter.

MUTATIONISM AND THE IMPACT OF MENDEL

- Gregor Mendel's research was published in 1866, but was not noticed until 1900.
- NOTE: Darwin knew nothing about the mechanism of inheritance when he conceived of natural selection.
- MUTATIONIST THEORIES (based on Mendel's work):
  - Emphasized the importance of DISCRETE VARIATION
  - T. H. Morgan -- the founder of *Drosophila* genetics.
  - R. Goldschmidt -- theory of "hopeful monsters"(saltation evolution)

Mendelian genetics disproved Lamarckian and blending inheritance theories. However, the emphasis on discrete variation caused a rift between mutation and natural selection that was initially damaging to the field.

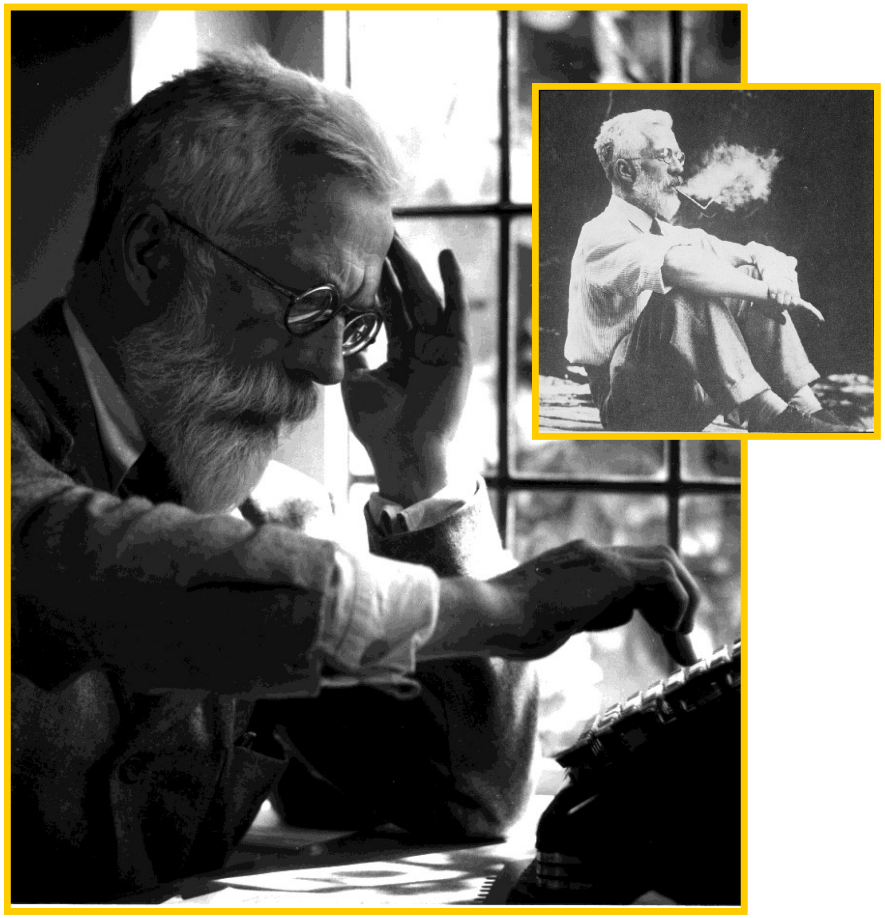
THE MODERN SYNTHESIS OF EVOLUTIONARY BIOLOGY

- The rift between genetics and natural selection was resolved in the 1930s and 1940s.
- This synthesis was forged from the contributions of geneticists, systematists, and paleontologists.

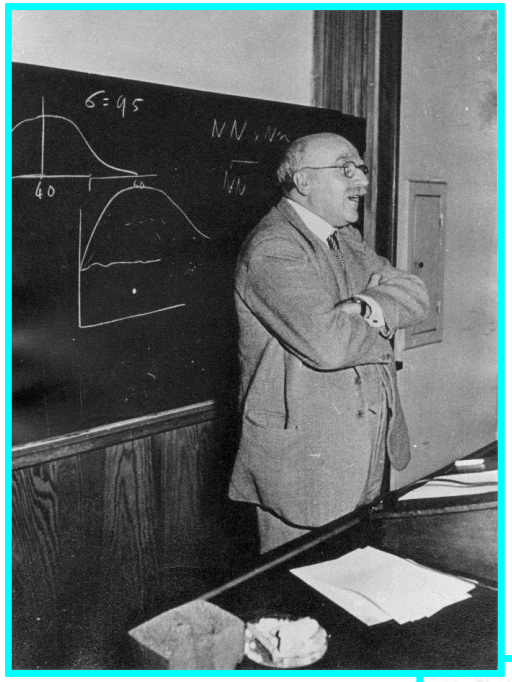
The Genetical Theory of Natural Selection
1930

- Fisher united Mendelian population genetics with the inheritance of continuous traits.

"The Correlation between Relatives on the Supposition of Mendelian Inheritance" *Philosophical Transactions of the Royal Society of Edinburgh* in 1918, (volume 52, pages 399–433).



Sir R. A. Fisher
1890-1962

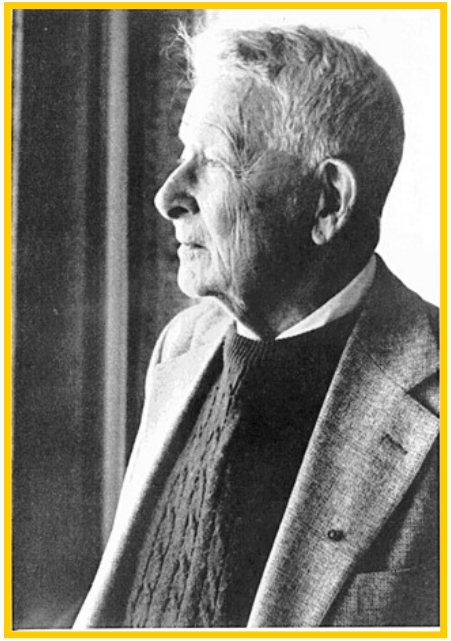


The Causes of Evolution
1932

- Developed the mathematical theory of gene frequency change under selection (and many other interesting applications).

J. B. S. Haldane
1892-1964

- Developed the mathematical framework for understanding the genetic consequences of migration, effective population size, population subdivision, and conceived of the concept of adaptive landscapes.



Sewall Wright
On his 90th birthday
1889-1988